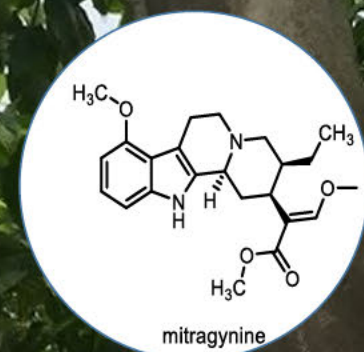


Kratom (*Mitragyna speciosa*): Recent understanding of leaf age and alkaloid content

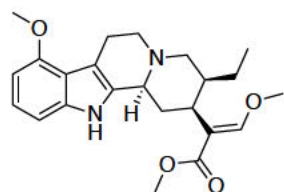
Christopher R. McCurdy, BScPh, PhD, FAAPS
The Frank A. Duckworth Eminent Scholar Chair
Professor of Medicinal Chemistry & Pharmaceutics
College of Pharmacy, University of Florida
Director, UF CTSI Translational Drug Development Core
AAPS President, 2017-2018



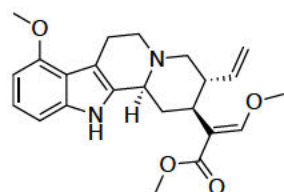
Mitragyna speciosa growing at the University of Florida MREC



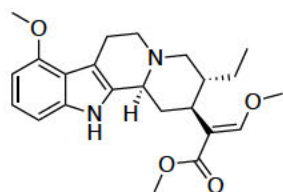
Isolation of kratom alkaloids



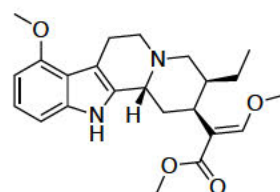
Mitragynine
66%
0.7-38.7%



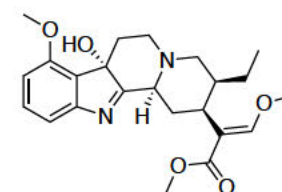
Paynantheine
9%



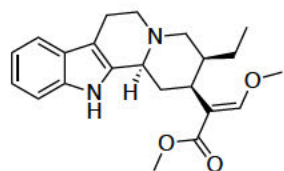
Speciogynine
7%



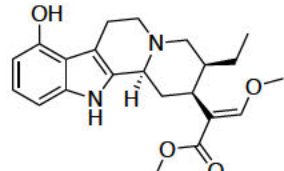
Speciociliatine
~1%



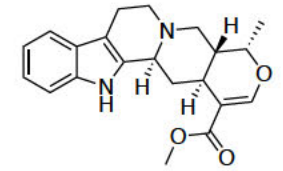
7α-hydroxymitragynine
~2%
>0.01%



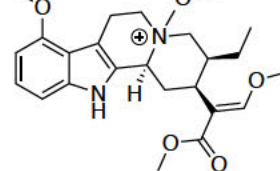
Corynantheidine
<1%



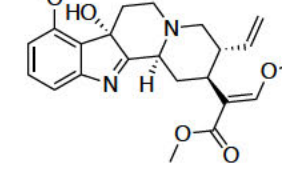
9-Hydroxycorynantheidine
<1%



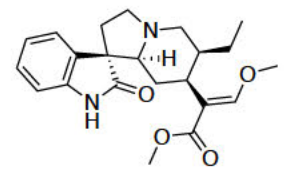
Ajmalicine
<1%



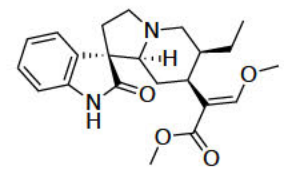
Mitragynine N-oxide
<1%



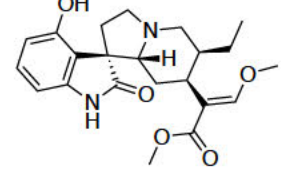
7α-hydroxypaynantheine



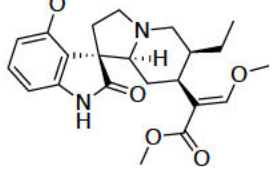
Corynoxine A
<1%



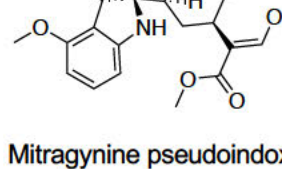
Corynoxine B
<1%



Isospeciofoline
<1%



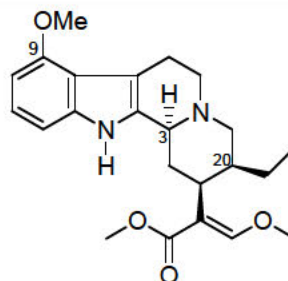
Mitragynine oxindole B
<1%



Mitragynine pseudoindoxyl

Mitragynine Eurofins screen at 82 CNS drug targets

Assay	100 nM (1.0E-07 M)	10000 nM (1.0E-05 M)
5-HT _{1A} (h) (agonist radioligand)		x
5-HT _{2B} (h) (agonist radioligand)		x
α_{1A} (h) (antagonist radioligand)		x
α_{1D} (h) (antagonist radioligand)		x
β_2 (h) (antagonist radioligand)		x
D ₁ (h) (antagonist radioligand)		x
D _{2S} (h) (agonist radioligand)		x
D ₃ (h) (antagonist radioligand)		x
κ (KOP) (agonist radioligand)		x
μ (MOP) (h) (agonist radioligand)		x
Na ⁺ channel (site 2) (antagonist radioligand)		x
Potassium Channel hERG (human)- [3H] Dofetilide		x

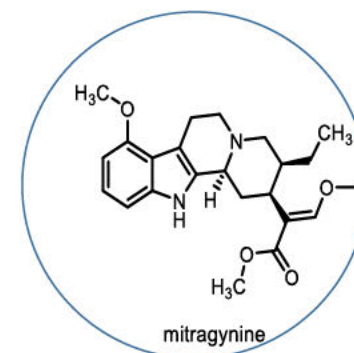


Mitragynine

Binding and functional effects of mitragynine

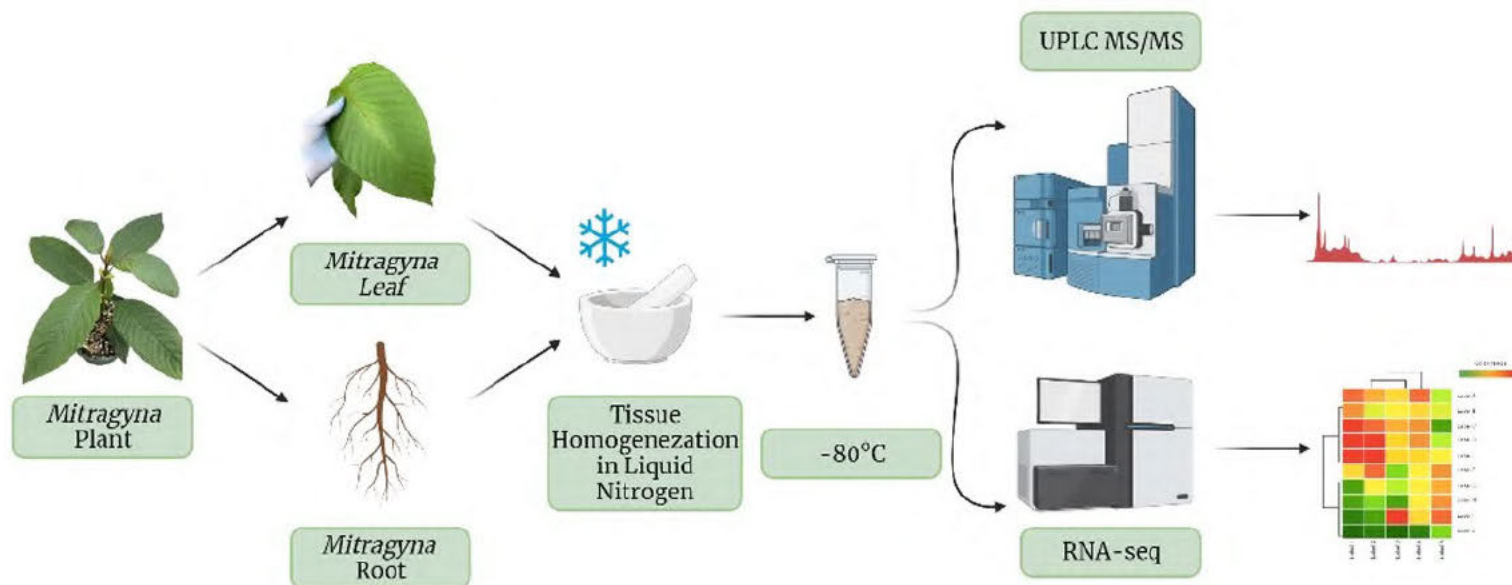
- Mitragynine has **partial agonist effects** at the μ opioid, α_{1A} and α_{1D} adrenergic receptors
- Poison Control Centers: Kratom overdoses resemble **stimulants** (not opioids)

Target	Ki (nM)	Agonist/ Antagonist
μ	136.0	Partial agonist
κ	157	Antagonist
δ		Antagonist
α_{1A}	1,660	Partial agonist
α_{1B}	2,490	Antagonist
α_{1D}	4,610	Partial agonist
α_{2A}	3,590	ND
α_{2B}	9,190	ND
α_{2C}	1,400	ND



Obeng, Samuel, et al. *J Med Chem.* 63 (2019). 433-439.

Defining the genetic basis of alkaloid biosynthesis in *Mitragyna speciosa*



- ❖ Integrate metabolomics and transcriptomics to define the genetic basis of alkaloid biosynthesis in *Mitragyna speciosa*

Discovery of central biosynthetic steps in MIA scaffold formation

JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY

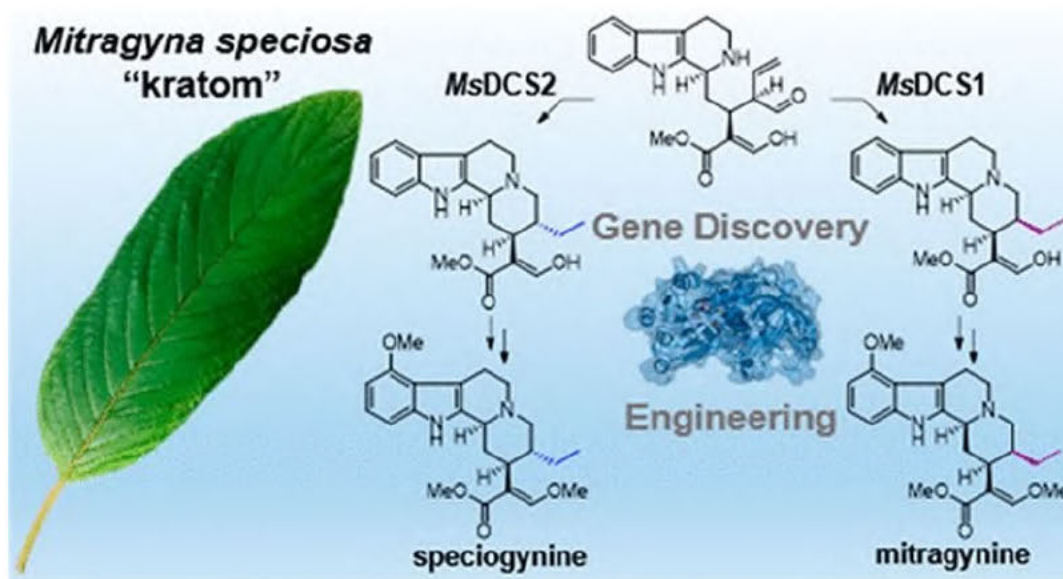
pubs.acs.org/JACS



Communication

Directed Biosynthesis of Mitragynine Stereoisomers

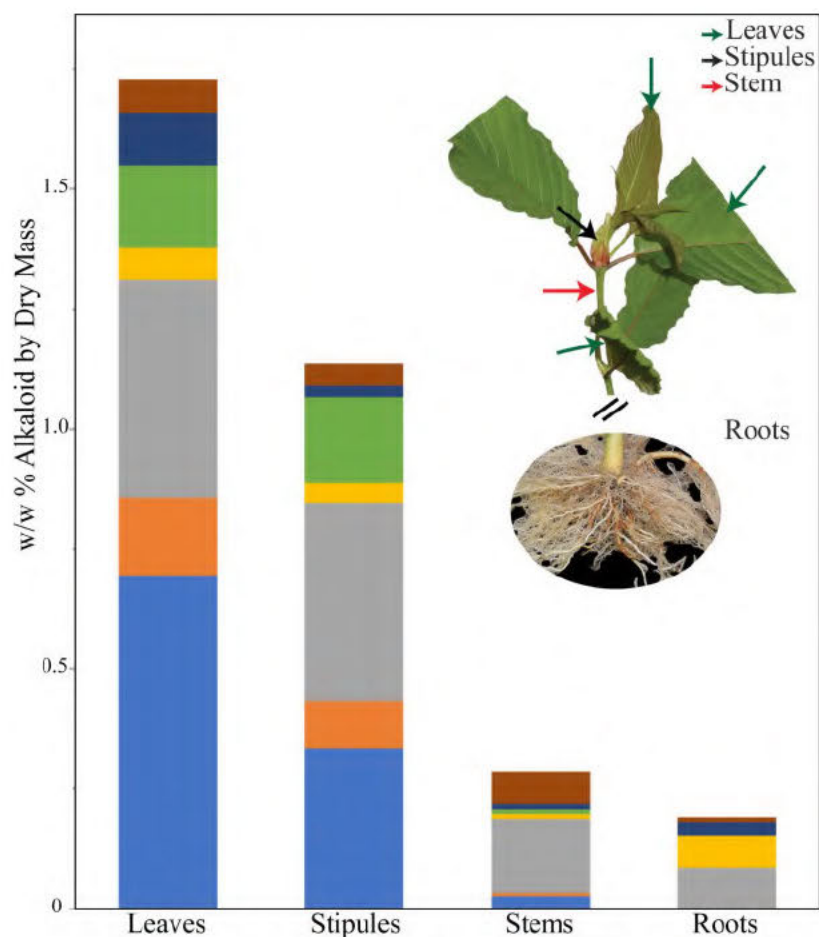
Carsten Schotte,¹ Yindi Jiang,¹ Dagny Grzech, Thu-Thuy T. Dang, Larissa C. Laforest, Francisco León, Marco Mottinelli, Satya Swathi Nadakuduti, Christopher R. McCurdy, and Sarah E. O'Connor*



J. Am. Chem. Soc. (2023) 145 9: 4957–4963

UF | College of Pharmacy
UNIVERSITY of FLORIDA

Characterization of alkaloids across tissue types in *M. speciosa*



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Article

Metabolite and Molecular Characterization of *Mitragyna speciosa* Identifies Developmental and Genotypic Effects on Monoterpene Indole and Oxindole Alkaloid Composition

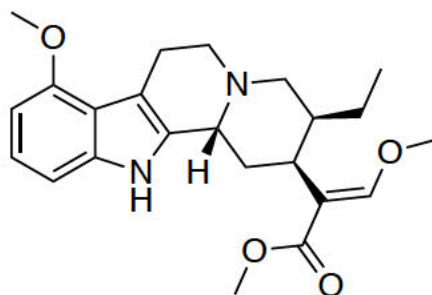
Larissa C. Laforest, Michelle A. Kuntz, Siva Rama Raju Kanumuri, Sushobhan Mukhopadhyay, Abhishek Sharma, Sarah E. O'Connor, Christopher R. McCurdy, and Satya Swathi Nadakuduti*

J Nat Prod (2023) 86, 4:1042–1052

❖ Mitragynine was the predominant alkaloid in leaves

Binding and function of speciociliatine at opioid receptors

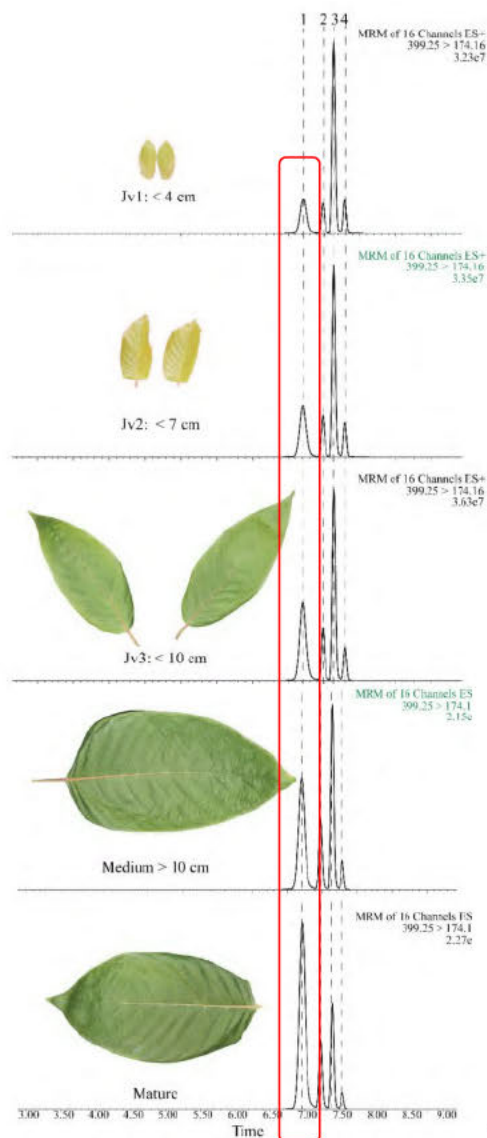
Target	Ki (nM)	Potency (nM)	Efficacy (% agonist response)	Agonist/ Antagonist
μ	39.8	39.2	73.6	Agonist
κ	98.1	No agonist effect	No agonist effect	No agonist or antagonist effects



Speciociliatine has a partial agonist effect at the MOR

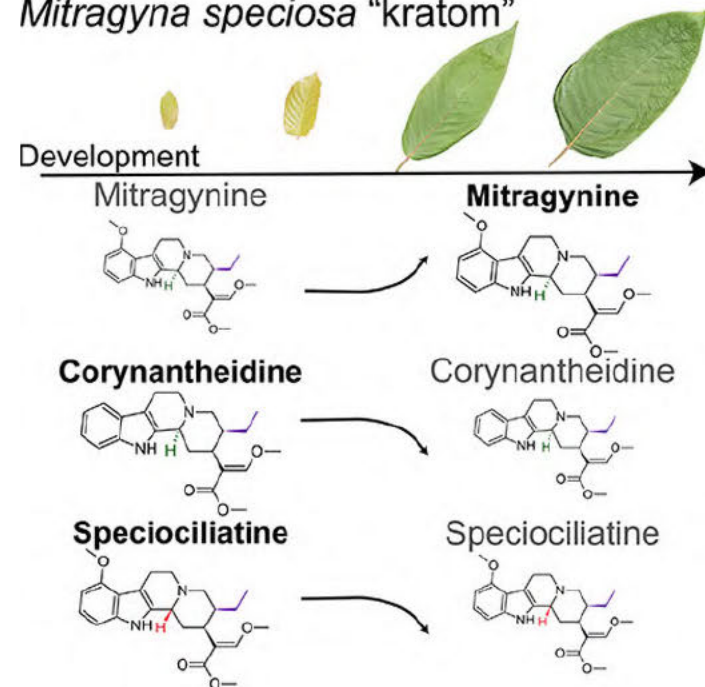
Obeng, Samuel, et al. *J Med Chem.* 63 (2019). 433-439.

Developmental alkaloid profiling of leaves



- (1) mitragynine,
- (2) speciogynine,
- (3) speciociliatine,
- (4) mitraciliatine

Mitragyna speciosa "kratom"



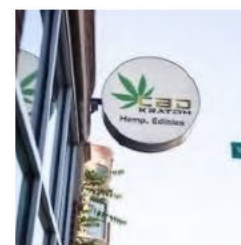
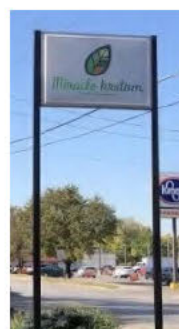
J Nat Prod (2023) 86, 4:1042–1052



Alkaloidal content in 211 commercial kratom products available in the USA



Product Content % (w/w%)										
Alkaloid	Mitragynine	7-Hydroxymitragynine	Speciogynine	Speciociliatine	Mitraciliatine	Mitraphylline	Paynantheine	Corynantheidine	Corynoxine	Corynoxine B
Range	0.27-1.72	BLLOQ-0.03	0.04-0.34	0.16-0.79	0.01-0.11	BLLOQ	0.05-0.45	0.01-0.07	BLLOQ-0.18	BLLOQ-0.07
Mean ± S.D.	0.79 ± 0.21	-	0.16 ± 0.05	0.33 ± 0.09	0.04 ± 0.01	-	0.21 ± 0.06	0.02 ± 0.01	-	



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- Francisco Leon, PhD
- Marco Mottinelli, PhD
- Sushobhan Mukhopadhyay, PhD
- Nelson Chear (USM)

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- Takato Hiranita, PhD
- Clinton Canal, PhD (Mercer University – 5HT)

- **DMPK**

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- Abhisheak Sharma, PhD
- Shyam Kamble, PhD
- Raju Kanumuri, PhD
- Tamara King, PhD
- Erin Berthold
- Michelle Kuntz

- **Behavioral Pharmacology**

- Lance McMahon, PhD
- Takato Hiranita, PhD
- Jay McLaughlin, PhD
- Jenny Wilkerson, PhD

- **Horticultural Science**

- Brian Pearson, PhD (UF/IFAS)
- Satya Swathi Nadakuduti (UF/IFAS)

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- NIDA R01 DA047855
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- Scott Akers and Jeremy Steding
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